

Southern Hemisphere ionosphere and plasmasphere response to the interplanetary shock event of 29–31 October 2003

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Abstract

[1] We analyze the effects on the Southern Hemisphere ionosphere and plasmasphere due to the 29–31 October 2003 geomagnetic storms (the so-called series of Halloween storms). Solar wind data from ACE and ionospheric data from the GPS (Global Position System) ground and LEO (Low Earth Orbit) receivers, the TOPEX/Poseidon altimeter, the IMAGE FUV camera, and the DMSP drift meter are used to understand the ionospheric dynamics as a function of the storm phase. The detailed structure of the ionosphere has been obtained using tomographic reconstruction applied to data from both ground- and space-based GPS receivers. The tomographic approach using LEO observations of signals received from GPS satellites above the LEO's horizon allows us to investigate the topside ionospheric and plasmaspheric density distribution in more detail than can be obtained using ground-based GPS receivers. This is because with ground-based receivers, the higher topside ionosphere and plasmasphere contribute only a small fraction to the total electron content (TEC) and so the measurements are dominated by the ionospheric structure at the $F2$ peak. In contrast, the Australian LEO satellite, FedSat, which has been used for this study, orbits at 800 km altitude, well above the $F2$ peak and hence the TEC measured is primarily due to the upper topside ionosphere and plasmasphere. This paper presents the tomographically reconstructed topside ionosphere and plasmasphere electron density distributions using LEO observations. The temporal and regional maps of TEC and the IMAGE FUV data show that the storm that commenced on 29 October dramatically decreased the plasma density in the Southern Hemisphere middle and high latitudes. The region remained depleted of plasma for more than 24 hours until 31 October, when the second severe storm began. TOPEX/Poseidon data shows a daytime localized density enhancement occurred above the middle of the Pacific Ocean. These results show large interhemispheric and longitudinally narrow storm-time structure in the ionosphere and topside ionosphere/plasmasphere.

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